Greecon
SH.P.K

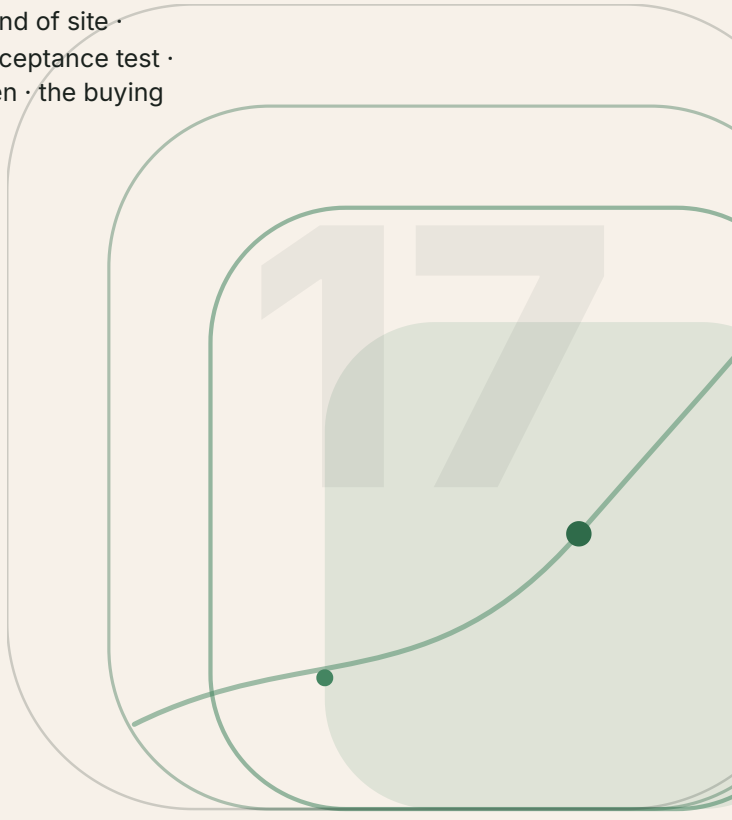
17

HARDWARE SHEET HW17 · QUALIFIED BY SPECIFICATION

Enclosure and fittings

The one box on the wall that holds the converter, the modules, the supply, the fuses and the terminals, with everything labelled.

QUALIFIES	Qualified by specification. Any enclosure of IP54 (IP65 outdoors) with a DIN rail, terminals, fuses or small breakers and glands, large enough for the kit.
USED IN	Kits H, S, F and W (H hotel or guesthouse, S solar plant, F farm or greenhouse, W water tank and pump station)
FOR	The project manager who buys and mounts it, and the electrician who connects its mains side
COVERS	How it qualifies · what it does at each kind of site · settings · fitting · in the platform · the acceptance test · what goes wrong · what is not yet proven · the buying record



Version 1.0 · 10 November 2026

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HW17

Enclosure and fittings

The enclosure is the one box on the wall that holds the kit: the fuses and the 24 V supply on its mains side; the converter, the modules and the terminal blocks on its low-voltage side; and the box itself, if there is room. It keeps dust and water out, it keeps mains and low voltage apart, and, labelled properly, it lets the next person understand the kit without asking anyone.

At a glance

WHAT IT DOES	Holds the converter, the modules, the supply, the fuses, the terminal blocks and, if there is room, the box.
HOW IT QUALIFIES	**Qualified by specification.** Any enclosure of IP54 (IP65 outdoors) with a DIN rail, terminals, fuses or small breakers and glands, large enough for the kit.
SPECIFICATION	IP54 (IP65 outdoors), DIN rail, terminals, fuses or small breakers for the meters' voltage taps, glands
FITTED BY	The project manager mounts it and the low-voltage parts in it. The electrician connects the mains side.
PLANNING ESTIMATE (EUR)	80 to 150. Not a quote.
USED IN	Kits H, S, F and W. The table under *At each kind of site* says what it does at each.

How it qualifies

Any make qualifies that meets every line below.

It qualifies if:

- ☐ It is rated **IP54** indoors, and **IP65** outdoors or where it can get wet.
- ☐ **It is large enough.** Lay the kit's parts side by side on a desk and measure: the DIN rail takes them all, with room for the terminal blocks and for air around the supply.
- ☐ It has a DIN rail, terminal blocks for the data cable and for 24 V, and fuses or small breakers: one for the supply's mains side and, where the electrician places them here, those of the meters' voltage taps.
- ☐ It has a cable gland for every cable that enters, at the bottom.
- ☐ Its door closes with a key or a tool.
- ☐ At a site with a 4G router: it is plastic, or the router's aerial can be led outside. A steel enclosure weakens the signal.

STOP

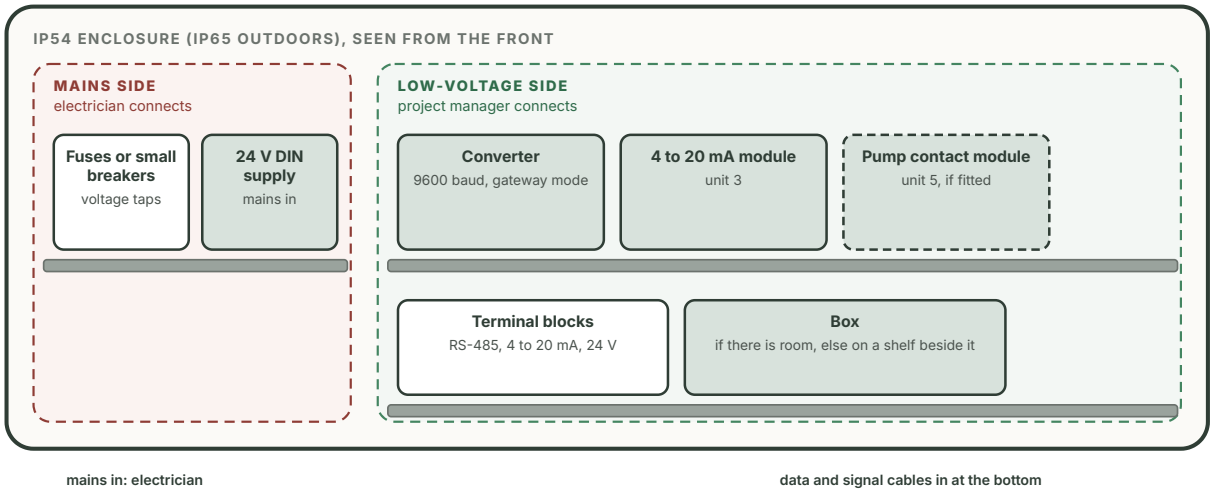
Mains and low voltage are kept apart in it, and the mains side is the electrician's. The project manager mounts the enclosure and connects the low-voltage side, and never connects, opens or alters the mains side.

At each kind of site

Kind of site	In its kit	How many	What it does there
H Hotel or guesthouse	IN THE KIT	1	One: the fuses and the 24 V supply on the mains side; the converter, the 4 to 20 mA module and the pump contact module on the low-voltage side; the terminal blocks and, if there is room, the box.
S Solar plant with grid connection	IN THE KIT	1	One, the smallest: the fuses and the supply, the converter, the terminal blocks and, if there is room, the box. The meters are in the distribution board, not here.
F Farm or greenhouse with irrigation	IN THE KIT	1	One, in the pump house: as at a hotel, with the 4G router on the rail. The probes and the weather sensor are in the field, on the data cable.
W Water tank and pump station	IN THE KIT	1	One: as at a hotel. The optional pump meter sits in the board that feeds the pump, not in this enclosure.

The order of the parts on the rail, for each kind of site. It is a recommendation, not a maker's drawing: the electrician may change it to suit the enclosure.

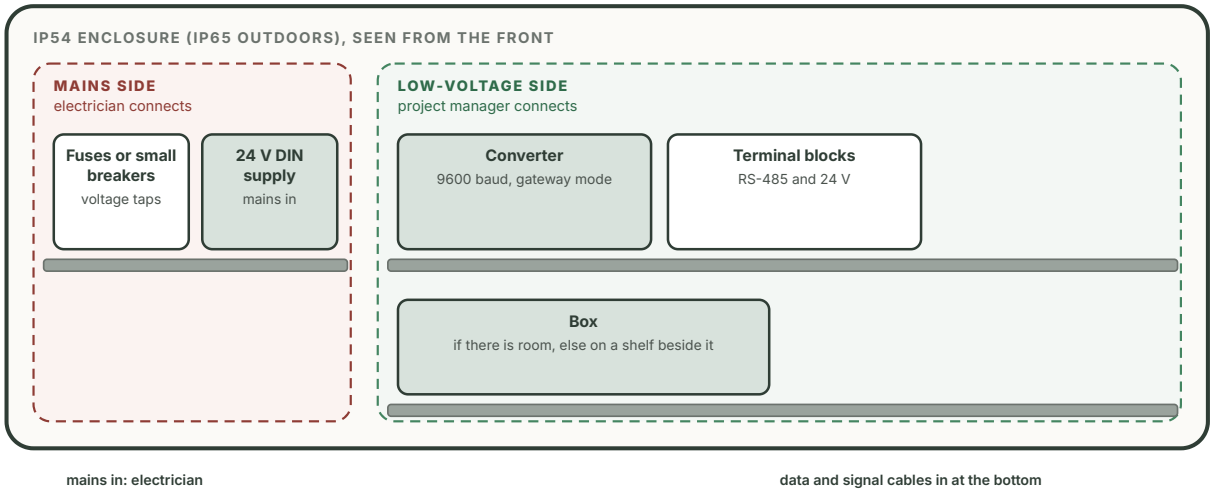
KIT H **The enclosure: recommended order on the DIN rail** ENCLOSURE



The meters are not in this enclosure: they sit in the distribution board at the point they measure.
The order is a recommendation, not a maker's drawing: the electrician may change it to suit the enclosure. Keep mains and low voltage apart, and label every wire at both ends. Terminal numbers are not drawn: follow the maker's manual for terminal numbers.

Figure H-4. Kit H. The enclosure: the recommended order of the parts on the DIN rail, mains side on the left.

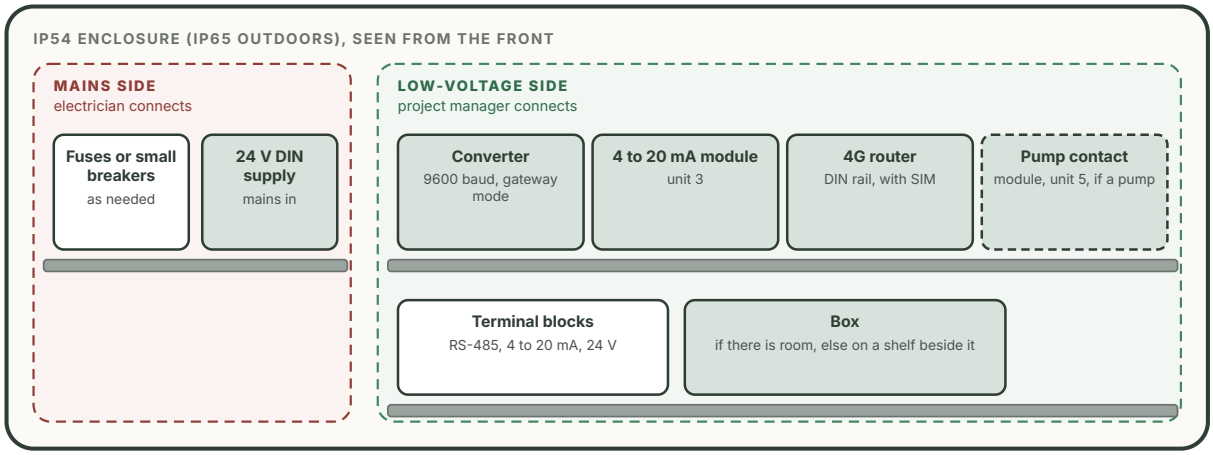
KIT S **The enclosure: recommended order on the DIN rail** ENCLOSURE



The meters are not in this enclosure: they sit in the distribution board at the point they measure.
The order is a recommendation, not a maker's drawing: the electrician may change it to suit the enclosure. Keep mains and low voltage apart, and label every wire at both ends. Terminal numbers are not drawn: follow the maker's manual for terminal numbers.

Figure S-4. Kit S. The enclosure: the recommended order of the parts on the DIN rail, mains side on the left.

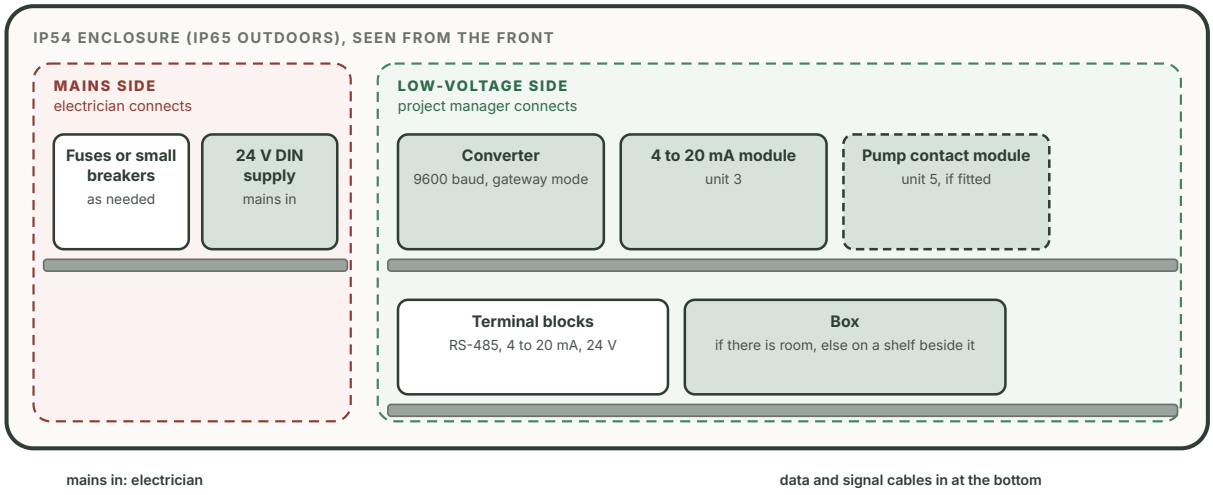
KIT F **The enclosure: recommended order on the DIN rail** ENCLOSURE



The probes and the weather sensor are in the field, on the data cable. Check the signal where the router sits.
The order is a recommendation, not a maker's drawing: the electrician may change it to suit the enclosure. Keep mains and low voltage apart, and label every wire at both ends. Terminal numbers are not drawn: follow the maker's manual for terminal numbers.

Figure F-4. Kit F. The enclosure: the recommended order of the parts on the DIN rail, mains side on the left.

KIT W **The enclosure: recommended order on the DIN rail** ENCLOSURE



The optional pump meter sits in the board that feeds the pump, not in this enclosure.
The order is a recommendation, not a maker's drawing: the electrician may change it to suit the enclosure. Keep mains and low voltage apart, and label every wire at both ends. Terminal numbers are not drawn: follow the maker's manual for terminal numbers.

Figure W-4. Kit W. The enclosure: the recommended order of the parts on the DIN rail, mains side on the left.

What is labelled

What	Its label
Each device	Its unit number, as set at the office
The converter	Its settings: gateway mode, 9600 baud, 8 data bits, no parity, 1 stop bit, port 502, and its address
Every wire	A number, at both ends
The UPS	The date of its battery
Inside the lid	The site name and the date

Never write a code or a password on the box, the enclosure or a label.

Fitting

PROJECT MANAGER mounts the enclosure and the low-voltage parts in it. ELECTRICIAN connects the mains side.

- ☐ **The place:** a dry wall, away from heat, where the door can be opened fully and the labels read. Outdoors, out of the sun where that is possible.
- ☐ **Cables come in at the bottom,** each through its own gland. Unused holes are plugged.
- ☐ **Mains on one side, low voltage on the other,** as the drawing shows, and the data cable away from power cables.

- [ ] The box goes inside if there is room, otherwise on a shelf beside it.
- [ ] **Photograph the inside before closing it,** with every label readable.

In the platform

The enclosure is not in the platform. What the platform keeps is what is written on its labels: each device's unit number and the converter's address, on the site's install card.

Acceptance test

At the office, before the kit leaves:

- [ ] The parts are mounted on the rail in the order of the drawing, and the door closes.
- [ ] Every device carries its unit number, and the converter its settings.
- [ ] The rehearsal on the desk was done with the parts in the enclosure, as they will be on site.

On the day, as part of commissioning:

Check and how	Pass when	Greecon shows	Real	Result
Closed up. Look before the door is shut.	Glands tight, unused holes plugged, mains and low voltage apart, every wire numbered at both ends, the labels in place, and the photograph taken			☐ Pass ☐ Fail

What goes wrong

What you see	Likely cause	What to do
Water or dust inside	A gland that was not tightened, an unused hole, or IP54 where IP65 was needed	Tighten and plug. Outdoors, or where it is hosed down, change it for IP65
The box restarts on hot days	The box and the supply in a small closed enclosure, in the sun or beside a heat source	Move the box to a shelf beside the enclosure, or move the enclosure into the shade
Nobody can tell which wire is which	The numbers were left off	Number both ends now, against the photograph, and take a new photograph
The 4G router hears nothing with the door shut	A steel enclosure	Lead the aerial outside, as the router's manual allows (sheet HW04)

Not yet proven

- **No kit has yet been assembled in an enclosure.** The order on the rail is a recommendation. The first one is assembled at the office and photographed, and this sheet is corrected from it.
- The prices are planning figures, not quotes.

Buying record

Ask three suppliers for a quote and keep the quotes in the shared drive. Buy one, pass the acceptance test with it, and only then buy more. For every part that is bought the hardware register keeps the make and model, the supplier, the date, the serial number or the batch, and the invoice.

	Quote 1	Quote 2	Quote 3
Supplier			
Make and model offered			
Price (EUR, without VAT), and delivery			
Meets every line of *It qualifies if*	☐ Yes	☐ Yes	☐ Yes

Bought	Answer
Make, model and version	
Supplier, date and invoice	
Serial number or batch	
Acceptance test passed on (date), by	